✅ DISCOVERY PHASE – COMPLETE

Locked Query: Review and optimize **Node B** (9060 XT 16 GB) which is currently running **Gemma-4-E4B Q8.**

Current Locked Roles (from previous decisions)

- **Node B**: Director / Apex Responder – fast general tasks, UI orchestration (Pretext HUD), medium synthesis, profile switching, swarm coordination.
- **Node C**: Ultra-fast / light tier (Gemma-4-E4B Q3 + Q4 clean).
- **Node D (K15):** Heavy Command Center (Gemma-4 26B Q6 clean + Qwen Coder 14B Q6 clean).

We are moving to **clean base models + Hyper-Tune weighting** across the board.

RESEARCH PHASE – Node B Hardware & Model Fit (9060 XT 16 GB)

Hardware Reality:

- 16 GB VRAM (AMD RDNA3) → excellent for high-quant models.
- Strong compute → can handle Q8 comfortably with good speed.

Gemma-4-E4B Q8 Current State:

- Fits easily (~12–13 GB with context).
- Very high coherence and tool-calling quality.
- However, it is still the **smaller E4B** variant (not the full 26B class).

Options for Node B:

1. **Keep Gemma-4-E4B Q8 (clean)** → Safe, fast, low risk.
2. **Upgrade to larger clean model** → Better Director intelligence.
3. **Hybrid / Specialist** → Add secondary model.

Recommendation Analysis:

Option	Model	Quant	VRAM Usage	Strengths for Node B Role	Headroom	Verdict

Keep current	Gemma-4-E4B	Q8	~12–13 GB	Fast, consistent, low latency	Excellent	Good but not maximal
Recommended	**Gemma-4 26B A4B** (clean)	**Q4_K_M**	~14–15 GB	Much stronger reasoning, better Director	~1–2 GB	**Best balance**
Aggressive	Gemma-4 26B	Q5_K_M	~18+ GB	Highest quality	Tight	Risky on 16 GB
Specialist	Qwen2.5-Coder-14B	Q6	~10 GB	Fast coding support	Excellent	Secondary only

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FINAL LOCKED RECOMMENDATION FOR NODE B

Upgrade Node B to Gemma-4 26B A4B Q4_K_M (clean base)

Reasons:

- 26B gives significantly better **Director-level intelligence** (planning, orchestration, profile handling, swarm synthesis).
- Q4_K_M fits comfortably in 16 GB with ~1–2 GB headroom (acceptable for Director role).
- Maintains family consistency with Node D's 26B Q6.
- Allows Hyper-Tune weighting / GEPA evolution to create a specialized "Director genome" on top of the clean base.
- Node C remains the true fast path (Q3/Q4), so Node B can afford to be a bit heavier.

Node B becomes:

- Primary Director (strong reasoning + orchestration)
- Secondary coding support via routing to Node D's Qwen Coder when needed

This creates a clean, powerful, non-redundant hierarchy:

- Node C → Fast path
- Node B → Director / medium-heavy

- Node D → Heavy brain + GEPA

Migration Plan (Clean Base + Hyper-Tune)

1. Download clean **Gemma-4 26B A4B** Q4_K_M.
2. Update model farm on Node B.
3. Run initial GEPA evolution pass focused on Director role (profile: "director").
4. Update router to prefer Node B for medium-orchestration tasks.

Would you like me to:

1. Give the **full Nix model-farm config** for Node B (Gemma-4 26B Q4 clean)
2. Provide the **router + profile updates** for the new hierarchy
3. Both + migration steps

Reply with 1, 2, or 3 and I'll deliver the exact implementation right now.

This upgrade makes Node B meaningfully stronger without breaking headroom or consistency. The entire cluster topology is becoming very clean and powerful. Your call?